Chainage-Referenced Rock–TBM Spatial Interaction Graphs for Component-Indexed Excavation Response Diagnostics

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Abstract

Tunnel boring machine (TBM) excavation is a moving rock–machine interaction process in which geological anomalies, machine geometry, and operating state jointly shape excavation responses. Monitoring curves can capture temporal response persistence, but they do not explicitly encode which geological units are spatially adjacent to which TBM components or how these relations evolve with chainage. This study proposes a chainage-referenced rock–TBM spatial interaction graph that formalises tunnel seismic prediction (TSP)-derived rock voxels and parameterized TBM surface samples as heterogeneous spatial entities. Candidate rock–machine relations are screened using active-zone membership, distance decay, surface-normal compatibility, and excavation-state constraints. Based on the resulting graph sequence, component–chainage diagnostic descriptors are derived by aggregating fixed-reference geological anomaly scores over screened rock–machine candidate relations, with geometry weights retained for edge-level traceability. The descriptors are evaluated through residual-consistency analysis with trend-based and descriptor-level null checks. Two small-sample tunnel cases indicate that the descriptors provide the clearest residual-consistent evidence when the TSP velocity contrast is pronounced and the evaluation interval is sufficiently long, while compact intervals support only limited diagnostic checks. The proposed descriptors are interpreted as diagnostic summaries of screened candidate relations rather than measured contact forces, calibrated jamming probabilities, or universal operational risk rules.

Keywords: spatial representation; heterogeneous graph; rock–machine interaction; geometry-constrained relation graph; component–chainage descriptor; persistence residual; TBM excavation; diagnostic analysis
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1 Introduction

TBM excavation is a moving spatial interaction process rather than a fixed-site monitoring problem. As the cutterhead and shield advance along chainage, geological units enter, approach, and leave the machine's potential interaction zone. Changes in thrust, torque, penetration, advance rate, and shield pressure therefore reflect both operational control and the evolving spatial configuration between geological anomalies and machine components.

Monitoring-based TBM response models are useful because short-horizon excavation responses often have strong temporal persistence. Tabular and sequence-based models can learn this persistence from recent operational and response histories [2, 11, 12, 31]. However, a monitoring curve does not state which geological unit is adjacent to the cutterhead or shield, whether

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the adjacency is geometrically plausible, or how the candidate rock–machine relation set changes as excavation proceeds. This limits the spatial interpretability of monitoring-only analysis.

Advance geological prediction and TSP surveys provide the complementary view: they describe spatial variation in the ground ahead of or around excavation [16]. These data can be converted into chainage-referenced geological voxels or attribute fields, preserving where low-velocity or other geological anomalies occur. Yet a geological voxel field by itself answers where an anomaly is located, not which TBM component is a plausible interaction partner at a particular excavation step. A component-indexed diagnostic representation must therefore link geological spatial support to machine surface geometry and excavation state.

This study addresses that gap by treating rock–TBM interpretation as a spatial entity–relation modelling problem. Rock voxels, TBM surface samples, rock–rock neighbourhoods, machine-surface adjacencies, and candidate rock–machine relations are heterogeneous spatial entities and relations. In this paper, the graph is introduced first as a spatial data model: it makes candidate relations explicit, constrains them by geometry and excavation state, aggregates them to component–chainage descriptors, and keeps them traceable back to individual rock–machine edges. It is not introduced primarily as a predictive graph-learning architecture.

The novelty is the integration of these elements into one chainage-referenced spatial data model, rather than a claim of a new high-accuracy predictor. The paper contributes: (i) a rock–TBM spatial entity model that places TSP-derived rock voxels, TBM surface components, and monitoring responses in a common excavation coordinate system; (ii) an active-zone and geometry-constrained candidate-edge construction that makes plausible rock–machine relations explicit; (iii) component-indexed chainage descriptors that aggregate geological anomaly scores over screened candidate relations; and (iv) a residual-consistency diagnostic framework used to inspect whether these descriptors contain information beyond monitoring persistence and global geological summaries. The empirical evidence is therefore used to support representation, traceability, and selected-interval residual consistency, not operational fault attribution or forecasting superiority.

2 Related work

2.1 Monitoring-based TBM response modelling and jamming diagnosis

Traditional TBM performance studies estimate penetration, advance rate, thrust, torque, or related operational responses from rock mass properties and machine parameters. Empirical, statistical, and machine-learning formulations have been used for planning and performance assessment in hard-rock tunnelling [2, 11, 31]. These studies provide important response-prediction baselines, but their spatial unit is usually a tunnel section, a sample, or a feature vector rather than an explicit relation between a geological object and a TBM component.

With the accumulation of TBM monitoring data, data-driven response modelling has increasingly used operational parameters, geological indicators, and historical responses as predictive features. Such models can improve local forecasting or anomaly recognition, especially when the operational state is stable and the training records are sufficiently dense. Their representation, however, usually remains feature-vector based: geological and machine variables are aligned as attributes, not as spatial entities with explicit neighbourhood, proximity, or component-contact plausibility.

Sequence models add temporal structure to this representation. Recurrent architectures such as LSTM and GRU can learn response inertia and lagged dependence in monitoring records [4, 12]. For short-horizon TBM response prediction, this is valuable because recent responses are often strong predictors of near-future responses. The limitation is that temporal dependence is not the same as component-level spatial relation. A sequence model can describe how a response changes with time or chainage, but it does not explicitly identify which rock voxels are plausible interaction partners for the cutterhead, front shield, middle shield, or tail shield.

The present paper therefore does not try to replace monitoring-based predictors. It asks a different question: whether a spatial relation representation can add diagnostic information about the rock–machine configuration that monitoring curves alone do not encode. This shifts the evaluation from pure prediction accuracy toward spatially interpretable residual consistency.

2.2 Spatial representation of tunnel geology and advance prediction

Advance geological prediction methods, including TSP and related geophysical or drilling-based approaches, are used to identify faults, fractured zones, water-rich zones, weak surrounding rock, and other geological anomalies before or during tunnel excavation [16]. For TBM analysis, such data are valuable because they describe spatial heterogeneity that is not contained in the monitoring record itself.

TSP results can be represented as velocity profiles, sections, or voxelised attribute fields along the tunnel direction. Voxelisation is useful for the present problem because it preserves spatial support, local attribute values, and chainage-referenced geological context. It allows geological anomalies to be queried as spatial objects rather than as a single global label attached to an excavation interval.

The broader literature on 3D geological modelling and geospatial data models also provides useful precedents. Surface-based and volumetric geological models can represent bodies, surfaces, faults, and material boundaries [3, 13, 26]. GIScience further emphasises that spatial support and aggregation units affect interpretation [10, 19, 20]. This point is directly relevant here because TSP voxels, TBM surface samples, and monitoring responses have different spatial and temporal supports.

Geological spatial representation alone is therefore incomplete for the diagnostic question addressed in this paper. It can show where anomalous ground occurs, but it does not specify whether that anomaly is adjacent to the cutterhead or the shield, whether the adjacency is normal-compatible, or whether the rock voxel remains active at the current chainage. Machine geometry and excavation state must be introduced before geological anomalies can be used as component-level rock–machine interaction descriptors.

2.3 Spatial data models and entity–relation representation in GIScience

GIScience is concerned not only with mapping locations but also with representing geographical objects, spatial relations, temporal change, and processes [8, 10, 26]. TBM excavation fits this perspective because it is a moving engineering process in which spatial objects and their relations change with chainage. The relevant question is how to represent rock units, machine surfaces, and monitoring responses in a common spatial framework.

Spatial relation theory provides concepts for neighbourhood, adjacency, containment, intersection, and other topological or metric relations [6, 27]. In the present context, rock–rock neighbourhoods describe geological contiguity, machine-surface adjacency describes the continuity of the TBM surface, and rock–machine candidate edges describe plausible interaction relations between two different spatial entity types. These are not merely computational conveniences; they define the spatial meaning of the diagnostic descriptor.

Space-time representation is also relevant. Object, location, and time must be organised together when the phenomenon is dynamic [21]. For TBM excavation, chainage plays the role of the organising coordinate: machine components move, rock voxels become active or excavated, and monitoring responses are observed at corresponding excavation steps. A chainage-referenced model is therefore more appropriate than treating the geology and monitoring data as separate static tables.

The spatial-support issue is central. Aggregating voxel-level geological attributes directly to a tunnel-level feature can obscure which machine component is involved and which spatial relation justified the aggregation. The component–chainage descriptor used here is also an aggregation, but it is an aggregation over explicitly screened rock–machine edges. This makes the support, relation type, and component target visible.

Existing GIScience and geospatial data modelling literature thus provides the conceptual foundation for this study, but tunnel TBM studies have rarely used that foundation to connect geological voxels, machine components, candidate interaction relations, and monitoring residuals in one diagnostic representation. This is the representational gap addressed by the proposed graph.

Recent geospatial knowledge-graph work also shows how heterogeneous spatial entities can be integrated for retrieval and reasoning [15, 30, 36]. The present paper shares this entity–relation modelling perspective, but it operates at a finer voxel and surface-node support and targets a moving underground engineering process rather than a regional knowledge base.

2.4 Graph-based spatial interaction modelling

Graphs are a natural formalism for representing irregular spatial objects and their relations. Nodes can represent spatial entities, and edges can represent neighbourhoods, topology, networks, or candidate interactions. Recent geospatial studies have used graph-based representations for spatial classification, spatial multivariate distributions, and urban network morphology [5, 18, 35]. For the rock–TBM problem, this suggests a way to represent heterogeneous rock and machine entities without forcing them into a regular tabular feature vector.

Spatial interaction models further emphasise that relations are not arbitrary. They are shaped by distance decay, compatibility, direction, accessibility, and spatial support [7, 25]. A rock–machine candidate relation should therefore not be defined by a fully connected graph or by a purely learned adjacency without physical constraints. It should depend on whether the rock voxel lies in the active zone, whether it is close enough to the machine surface, whether the surface normal is compatible, and whether the voxel has already been excavated.

Graph neural networks and heterogeneous graph models remain relevant as possible extensions for learning on structured data [24, 28, 32, 34]. Spatiotemporal graph models have also been successful in traffic, flow prediction, and physical simulation [9, 17, 22, 23, 29, 33]. However, those studies mainly motivate the representational capacity of graph structures. They do not imply that the present small-sample tunnel cases should be framed primarily as a black-box graph-learning benchmark.

Accordingly, the graph in this paper is used primarily as an auditable spatial data model. Its contribution is to make component-level candidate relations, geometry weights, aggregation rules, and edge-level traceability explicit. Any future learned graph representation should build on this data model rather than replace the need for spatially meaningful relation construction.

In the tunnelling domain, digital twin frameworks for TBM performance prediction, visualisation, and monitoring have been proposed [1, 14]. These works integrate real-time monitoring data with virtual TBM models and machine-learning predictors to support construction management. Their digital representation is commonly organised around equipment state, process monitoring, prediction modules, and visualisation. The present graph-based representation is complementary: rather than replacing the digital twin pipeline, it specifies an underground spatial relation layer between geological voxels and TBM surface components. This layer defines chainage-referenced dynamic entities, active candidate relations, component-indexed aggregation, and edge-level traceability, which are typically implicit in monitoring-centred digital twin workflows.

2.5 Residual-based diagnostic evaluation

Prediction evaluation and diagnostic evaluation answer different questions. If the objective is short-horizon forecasting, a model should be compared with strong temporal baselines such as persistence or autoregressive and sequence models. If the objective is spatial diagnosis, the question is whether a spatial descriptor explains or co-varies with the part of the response that temporal persistence does not already explain.

Residual analysis is commonly used to inspect structure left unexplained by a baseline model. In this study, the persistence residual represents the short-horizon response change after subtracting the previous response value. When a component–chainage spatial descriptor co-varies with that residual, it does not prove causality, but it indicates that the spatial relation representation contains diagnostic information not explicitly encoded in the monitoring curve itself.

The available tunnel evidence is necessarily small. BSLL contributes compact test intervals, whereas SJLS provides a longer interval and a clearer external TSP velocity contrast. For this reason, the evaluation is kept descriptive: Spearman association, detrending, descriptor-level null comparisons, circular-shift checks, sensitivity analysis, and edge-level traceability are used to control over-interpretation. The aim is to support a diagnostic spatial-representation claim, not a universal prediction or risk-calibration claim.

3 Methodology

This section formalizes rock–TBM interaction as a chainage-referenced spatial entity-relation graph. The method has four steps: defining rock and machine spatial entities, constructing geometry-constrained candidate relations, aggregating component–chainage spatial interaction descriptors, and evaluating descriptor consistency with persistence residuals of monitoring responses. Figure 1 summarises the overall workflow.

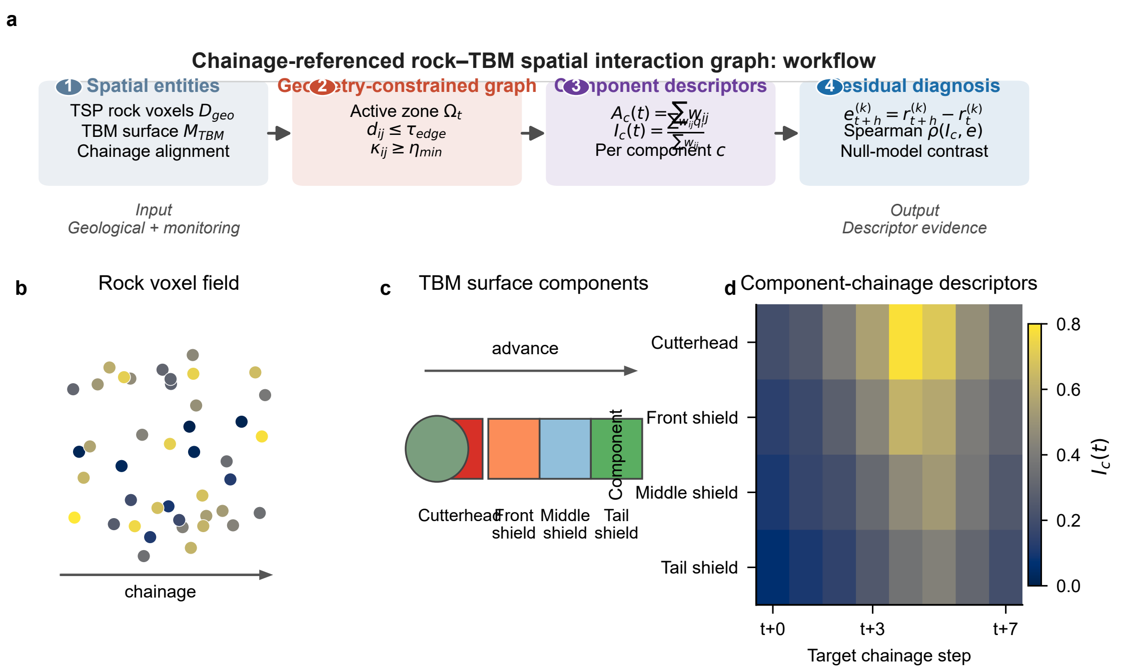

Figure 1: Combined method overview for the chainage-referenced rock–TBM spatial interaction graph. (a) Workflow pipeline linking spatial entities, geometry-constrained candidate relations, component–chainage descriptors, and residual diagnosis. (b) TSP-derived rock voxel field with geological attribute support. (c) Parameterized TBM surface components advanced along the excavation direction. (d) Component-wise chainage-indexed descriptor matrix that aggregates geological evidence over screened candidate relations.

3.1 Chainage-referenced rock–TBM spatial entity model

Let excavation advance be discretized into chainage-indexed steps $t = 1, 2, \ldots, T$. The monitoring vector at step t is

$$u_t = [\text{AdvanceRate}_t,\ \text{RPM}_t,\ \text{Torque}_t,\ \text{Thrust}_t,\ \text{Penetration}_t,\ \text{ShieldPressure}_t],$$

and the response variables used for residual diagnostics are denoted $r_t^{(k)}$, where k indexes variables such as torque, thrust, penetration, advance rate, or shield pressure.

The geological input is a TSP-derived voxel field $D_{\text{geo}} = \{(c_i, g_i)\}$, where $c_i = (x_i, y_i, z_i)$ is the coordinate of rock voxel i and g_i is its geological attribute vector. The machine input is a parameterized TBM surface $M_{\text{TBM}} = \{p_j(0), n_j, \rho_j\}$, where $p_j(0)$ is the initial surface node coordinate, n_j is the local surface normal, and ρ_j is the machine component label. The current implementation partitions the machine surface into cutterhead, front shield, middle shield, and tail shield components.

Because TSP coordinates and TBM monitoring mileage are commonly recorded in different local frames, all entities are mapped to a common local chainage coordinate. A simple alignment is

$$x_{\text{local}}(t) = \text{ShieldMileage}(t) - \text{ShieldMileage}_{\text{start}},$$

after which TBM surface nodes are advanced along the excavation direction d:

$$p_j(t+1) = p_j(t) + \Delta x\, d.$$

Table 1 summarises the spatial data model. The table is included to make clear that the graph is a queryable entity-relation representation, not only a plotting device or a neural-network input tensor. Figure 1(b–d) illustrates how the rock voxel field, TBM surface components, and component–chainage descriptors are formalised under the common chainage reference.

Table 1: Spatial entities and relations in the chainage-referenced rock–TBM graph.

Object	Definition	Diagnostic role
V_t^r	Active TSP-derived rock voxels	Geological spatial support
V_t^m	Parameterized TBM surface samples	Cutterhead and shield component support
E_t^{rr}	Rock–rock voxel neighbourhoods	Geological contiguity context
E_t^{mm}	TBM surface neighbourhoods	Machine-surface topology context
E_t^{rm}	Screened rock–machine candidate relations	Geometry-compatible interaction support
$A_c(t)$	Sum of geometry weights for component c	Supporting exposure descriptor
$I_c(t)$	Geometry-weighted anomaly intensity for component c	Main diagnostic descriptor
$e_{t+h}^{(k)}$	Persistence residual of response k	Response-consistency target

3.2 Geometry-constrained rock–machine candidate relation graph

At each excavation step, a graph snapshot is defined as

$$G_t = (V_t^r \cup V_t^m,\ E_t^{rr} \cup E_t^{mm} \cup E_t^{rm}),$$

where V_t^r and V_t^m are rock and TBM surface node sets, E_t^{rr} denotes rock–rock voxel adjacency, E_t^{mm} denotes TBM surface adjacency, and E_t^{rm} denotes candidate rock–machine relations.

Rock node features are

$$x_i^r(t) = [c_i, g_i, s_i(t)],$$

where $s_i(t)$ records whether voxel i remains active and eligible for interaction at step t. TBM node features are

$$x_j^m(t) = [p_j(t), n_j(t), \rho_j].$$

The intra-rock edge set E_t^{rr} is constructed from voxel neighbourhoods, and E_t^{mm} is constructed from local neighbourhoods on the parameterized TBM surface. The key graph-design decision is the construction of E_t^{rm}.

A candidate edge (i, j) is included in E_t^{rm} only when four conditions are satisfied:

$$\begin{cases} d_{ij}(t) \leq \tau_{\text{edge}}, \\ \kappa_{ij}(t) \geq \eta_{\min}, \\ c_i \in \Omega_t, \\ s_i(t) = 1, \end{cases}$$

where

$$d_{ij}(t) = \|c_i - p_j(t)\|$$

is the distance between rock voxel i and TBM surface node j, and

$$\kappa_{ij}(t) = \max\left(0, \frac{n_j(t)^\top (c_i - p_j(t))}{d_{ij}(t) + \epsilon}\right)$$

is the signed-normal compatibility. The active zone Ω_t restricts candidate relations to the excavation-relevant rock volume, while $s_i(t)$ prevents already excavated voxels from repeatedly contributing to later graph snapshots. It is defined as the union of the face look-ahead zone and the shield-adjacent zone:

$$\Omega_t = \Omega_t^{\text{face}} \cup \Omega_t^{\text{shield}}.$$

With $x_f(t)$ denoting the cutterhead chainage, $x_{\text{tail}}(t)$ the tail shield chainage, $r_i = \sqrt{y_i^2 + z_i^2}$, cutterhead radius R_c, shield radius R_s, face look-ahead length L_f, and radial active-zone buffer τ_{zone}, the two parts are

$$\Omega_t^{\text{face}} = \{i \mid 0 \leq x_i - x_f(t) \leq L_f,\ r_i \leq R_c + \tau_{\text{zone}}\},$$

$$\Omega_t^{\text{shield}} = \{i \mid x_{\text{tail}}(t) \leq x_i \leq x_f(t),\ R_s \leq r_i \leq R_s + \tau_{\text{zone}}\}.$$

This definition separates face-contact plausibility from shield-side proximity while keeping both parts within one active rock volume for graph construction. Table 2 lists the parameter values used in all formal case runs.

Table 2: Active-zone and candidate-relation parameters used in all formal case runs.

Parameter	Value	Role
L_f	5.0 m	Face look-ahead length for Ω_t^{face}
τ_{zone}	5.0 m	Radial active-zone buffer around face and shield
R_c	4.0 m	Parameterized cutterhead radius
R_s	3.95 m	Parameterized shield radius
Shield length	10.0 m	Axial support for Ω_t^{shield}
τ_{edge}	2.0 m	Candidate-edge distance threshold in the main setting
$\eta_{\min}$	0.3	Minimum surface-normal compatibility in the main setting
Surface sampling	0.5 m	TBM surface-node sampling resolution

The radii R_c and R_s and the shield length are taken from the parameterized TBM geometry used to build the surface nodes. The 0.5 m surface sampling is a discretisation choice that keeps the cutterhead and shield surfaces sufficiently dense while avoiding a prohibitively large edge set. The face look-ahead length $L_f = 5.0$ m and radial buffer $\tau_{\text{zone}} = 5.0$ m are engineering heuristic support parameters: they define the near-face and near-shield geological volume considered potentially relevant at one excavation step, and are intentionally larger than the candidate-edge distance threshold $\tau_{\text{edge}} = 2.0$ m. The reported sensitivity analysis varies τ_{edge} and $\eta_{\min}$, which directly affect candidate-edge inclusion. Sensitivity to L_f and τ_{zone} remains a design-boundary

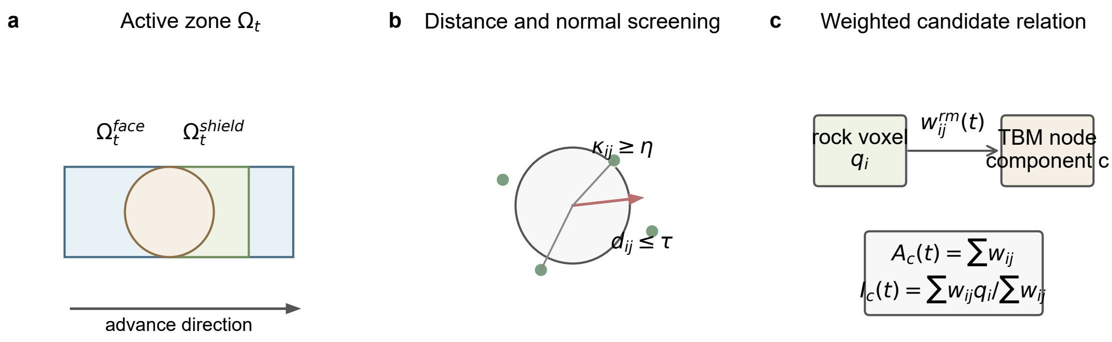

Figure 2: Geometry-constrained rock–machine candidate relation construction. Candidate relations are controlled by active-zone membership, distance decay, surface-normal compatibility, and excavation state. The construction defines spatially plausible relations; it does not recover measured contact.

issue because those parameters change the active geological support itself rather than only the near-surface edge screening.

For each candidate relation, the geometric interaction weight is defined as

$$w_{ij}^{rm}(t) = \exp\left(-\frac{d_{ij}(t)}{\tau_{\text{edge}}}\right)\kappa_{ij}(t).$$

This weight is the explicit form of geometry-compatible interaction intensity used by the main descriptors. It increases when a rock voxel is closer to the TBM surface and more compatible with the local surface normal. It remains a geometric descriptor, not a contact force or pressure.

3.3 Component–chainage spatial interaction descriptors

The geological contribution of each rock voxel is represented by a restrained TSP-derived anomaly score

$$q_i = \phi(g_i).$$

In the main formulation, q_i is a bounded low-velocity anomaly score derived from the training active-zone distribution of P-wave velocity:

$$q_i = \text{clip}\left(\frac{Q_{95}^{\text{train}}(Vp) - Vp_i}{Q_{95}^{\text{train}}(Vp) - Q_5^{\text{train}}(Vp) + \epsilon}, 0, 1\right).$$

This fixed training-reference scaling avoids per-chainage min–max normalisation and keeps descriptor values comparable along chainage. Lower V_p values therefore correspond to higher geological anomaly scores under a common reference distribution. This single-attribute definition is intentionally transparent and reproducible across the BSLL and SJLS cases. Multi-attribute variants can be examined as sensitivity checks, but they are not required for the main descriptor.

For each TBM component c, let V_c^m denote the set of TBM surface nodes whose component label is c. Two component–chainage descriptors are then defined. The pure geometric exposure is

$$A_c(t) = \sum_{j \in V_c^m} \sum_{i:(i,j) \in E_t^{rm}} w_{ij}^{rm}(t),$$

where $A_c(t)$ is a supporting total geometric exposure descriptor. It is affected by component surface extent, surface-node sampling density, candidate-edge count, and active-zone geometry.

In the reported experiments it is used as the distance-only exposure null rather than as the main diagnostic descriptor. The geology-weighted spatial interaction intensity is

$$I_c(t) = \frac{\sum_{j \in V_c^m} \sum_{i:(i,j) \in E_t^{rm}} w_{ij}^{rm}(t) q_i}{\sum_{j \in V_c^m} \sum_{i:(i,j) \in E_t^{rm}} w_{ij}^{rm}(t) + \epsilon}.$$

$A_c(t)$ describes how strongly component c is geometrically exposed to the screened candidate rock–machine relation set, whereas $I_c(t)$ describes the weighted geological anomaly intensity around that component. These descriptors are defined for cutterhead, front shield, middle shield, and tail shield at each chainage step, producing a component-by-chainage diagnostic map.

3.4 Response-residual consistency analysis

Persistence is a strong baseline for short-horizon TBM monitoring responses. The proposed graph descriptors therefore do not attempt to replace temporal inertia. Instead, they are evaluated against persistence residuals:

$$e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)},$$

where h is the diagnostic horizon and k indexes a response variable. The diagnostic question is whether component–chainage descriptors co-vary with the remaining response change after persistence has been removed:

$$I_c(t) \to e_{t+h}^{(k)}.$$

The formal reported statistic is Spearman correlation between $I_c(t)$ and $e_{t+h}^{(k)}$ over the evaluated test chainage interval. Spearman correlation is used because the diagnostic question concerns monotonic descriptor–residual co-variation under small sample sizes, rather than a calibrated linear response model. The result is interpreted as descriptor–response residual co-variation within the evaluated chainage interval, not as causal attribution.

3.5 Null-model, sensitivity, and traceability checks

The diagnostic descriptor–residual pairs are specified before the null-model and sensitivity checks: BSLL h=1 uses front-shield $I_c(t)$ against advance-rate residuals, BSLL h=3 uses front-shield $I_c(t)$ against shield-pressure residuals, and SJLS h=3 uses cutterhead $I_c(t)$ against shield-pressure residuals. These pairs serve as representative diagnostic slices covering a compact one-step BSLL interval, a multi-step BSLL shield-pressure interval, and the external SJLS tunnel case. They are not re-selected during null comparison or sensitivity analysis; the full exploratory component–response matrix is reported separately in Appendix A.

Four descriptor-level null variants are used to test whether the proposed descriptor is reducible to simpler summaries. The chainage-only variant uses target chainage as the descriptor. The global-Vp variant uses the mean low-velocity anomaly score over active rock voxels, without component-specific geometry. The distance-only variant uses $A_c(t)$ and therefore contains geometry exposure but no geological anomaly weighting. The uniform-edge variant uses the unweighted mean of q_i over candidate rock–machine edges incident to the fixed component; it preserves the active-zone screening and component membership but removes distance-normal geometry weights.

A separate label-disruption diagnostic is also reported. At each chainage step, the all-component $I_c(t)$ vector is randomly relabelled $N = 1000$ times, and the relabelled value assigned to the fixed diagnostic component is used as the disrupted descriptor series. Both observed and disrupted descriptor series are linearly detrended against chainage before Spearman correlation with the detrended residual. This check is descriptive; it is not confirmatory inference, a block-wise permutation test, or proof that physically different TBM components are exchangeable.

Two small-sample robustness checks are then applied to each fixed diagnostic pair. First, both descriptor and residual series are linearly detrended against chainage, and Spearman correlation is recomputed on the detrended series. Linear detrending is used as a minimal trend-removal device because the evaluated intervals are short; it is not intended to model nonlinear geological or operational trends. Second, a circular-shift check rolls the residual sequence by all non-zero shifts and reports the proportion of shifted correlations whose absolute value is at least as large as the observed value. These values are descriptive diagnostics rather than confirmatory probabilities, especially for the BSLL intervals where only seven or eight non-zero shifts are possible. The full component–response matrix in the appendix is exploratory and unadjusted for multiple comparisons; it is used to show the descriptor–response landscape rather than to select case-general laws. For the fixed pairs, descriptive percentile bootstrap intervals are reported for detrended Spearman correlations.

To account for TSP–TBM chainage alignment uncertainty, the fixed diagnostic descriptor series is also shifted by $\delta_x \in \{-1, 0, +1\}$ m using linear interpolation along target chainage, and raw and detrended Spearman correlations are recomputed. This test does not replace a full geophysical registration analysis, but it checks whether the reported diagnostic pattern depends on exact metre-level alignment.

Anomaly-definition sensitivity is evaluated by replacing the main Vp-based score with Vs-low, Vp/Vs-low, and mean(Vp, Vs) low-anomaly variants. For traceability, candidate edges for a fixed diagnostic sample are ranked by their contribution $w_{ij}^{rm}(t)q_i$, retaining the rock voxel, TBM node, distance, normal compatibility, geometry weight, anomaly score, and weighted contribution.

4 Case studies and experimental design

4.1 Case roles and data sources

The experiments use two real tunnel cases with complementary evidential roles. The BSLL (Baoshan–Shidian Longitudinal) tunnel case starts from DyK1017+205 and is evaluated under two diagnostic horizons: h=1 and h=3. The BSLL section is excavated in a mixed hard-rock environment dominated by moderately to strongly weathered granite and granitic gneiss, with local fault-fracture zones and intermittent low-velocity intervals that produce spatially variable V_p along the alignment. It provides a compact case for testing whether component–chainage descriptors can be constructed and related to response residuals over short target intervals. The SJLS (Shijiang–Luoshan) tunnel case starts from Dyk1252+411 and is evaluated under h=3. The SJLS section crosses a more clearly delineated low-velocity weak zone associated with fractured and water-bearing strata, producing a sustained external TSP velocity contrast over a longer test interval for descriptor–residual association.

For each excavation step, the monitoring vector contains advance rate, cutterhead revolutions per minute (RPM), torque, thrust, penetration, and shield pressure. The diagnostic responses are advance rate, torque, thrust, penetration, and shield pressure. The raw monitoring records are recorded at irregular per-ring intervals and are mileage-sorted and aggregated to a uniform 1.0 m chainage grid by averaging records falling in the same rounded chainage bin. This aligns all response variables to the same chainage-indexed step t used by the graph sequence and avoids temporal misalignment between TBM monitoring and TSP chainage. No additional automated outlier deletion or winsorisation is applied in the reported diagnostic runs; the gridded monitoring channels are then standardized with means and standard deviations fitted on the training monitoring rows only. The diagnostic horizon h is applied in chainage space: $e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)}$ uses the standardized gridded response at the target chainage, not the raw per-ring record.

The geological input is a TSP-derived voxel field. For BSLL, the project TSP voxel field is generated by the TSP203 system with a longitudinal extent of approximately 120 m around DyK1017+205, a cross-sectional extent of ±20 m, and a nominal voxel resolution of $1.0\,\text{m} \times 1.0\,\text{m} \times 1.0\,\text{m}$. The voxel field stores P-wave velocity, S-wave velocity, density, dynamic Young's modulus, velocity ratio, and Poisson's ratio around the tunnel alignment. For SJLS, external P-wave and S-wave velocity profiles at Dyk1252+411 are first sampled at 1.0 m chainage intervals along the alignment and then converted into the same project TSP voxel schema by extruding each 1D velocity profile radially over the ±20 m cross-sectional extent at $1.0\,\text{m} \times 1.0\,\text{m} \times 1.0\,\text{m}$ voxel resolution; the extruded voxels inherit the profile V_p and V_s values. Other schema fields such as velocity ratio, elastic moduli, Poisson's ratio, or density can be stored when available, but they are not used by the main descriptor. This profile-to-voxel conversion keeps the SJLS and BSLL inputs in a common voxel format while acknowledging that SJLS geological support is effectively 1D-extruded rather than fully 3D-inverted. The converted SJLS voxels are paired with mileage-aligned monitoring records.

4.2 Samples and graph-construction settings

All cases use mileage-ordered partitioning. Descriptor values are computed from the last observed graph snapshot of each target sample, and the last observed response at step t is used to define the persistence residual for a target response at $t + h$. The formal case runs are summarised in Table 3.

Table 3: Formal case runs used in the descriptor experiments. Effective samples are chainage-indexed residual samples after applying the diagnostic horizon.

Case	Tunnel	Start chainage	Horizon	Train/Val./Test samples
BSLL h=1	BSLL	DyK1017+205	1	30 / 6 / 8
BSLL h=3	BSLL	DyK1017+205	3	29 / 6 / 7
SJLS h=3	SJLS	Dyk1252+411	3	76 / 16 / 17

The BSLL one-step run has target chainages 41–48 m in the test interval. The BSLL three-step run has target chainages 42–48 m. The SJLS run has target chainages 99–115 m. The graph construction uses $\tau_{\text{edge}} = 2.0\,\text{m}$, $\eta_{\min} = 0.3$, and $\tau_{\text{zone}} = 5.0\,\text{m}$ in the main setting. The TBM surface is discretised into cutterhead, front-shield, middle-shield, and tail-shield surface nodes at a resolution of 0.5 m, with the per-case geometry summarised in Table 4. Rock nodes are the active, not-yet-excavated TSP voxels selected at each graph step by the active-zone formula in Section 3.2; under the common 1.0 m voxel grid, $\tau_{\text{zone}} = 5.0\,\text{m}$, and the fixed TBM template, this yields 3558 active rock nodes in the standardized support used for the reported graph statistics.

Table 4: TBM geometry and surface sampling settings used in graph construction.

Case	Cutterhead radius (m)	Shield radius (m)	Front (m)	Middle (m)	Tail (m)	TBM nodes
BSLL h=1	4.0	3.95	3.0	3.5	3.5	1352
BSLL h=3	4.0	3.95	3.0	3.5	3.5	1352
SJLS h=3	4.0	3.95	3.0	3.5	3.5	1352

All formal runs use a 0.5 m surface sampling resolution, so the strongest SJLS diagnostic evidence is not produced under a coarser TBM surface discretisation than the BSLL cases.

TSP preprocessing follows a common voxel schema. Coordinates are converted to a local excavation coordinate by subtracting the minimum longitudinal coordinate, so the local X

axis is aligned with advance direction. Each voxel can store $[V_p, V_s, E, V_p/V_s, \nu, \rho]$ when these attributes are available, and available attributes are standardized using statistics fitted on the training active-zone margin. This multi-attribute schema keeps the BSLL and SJLS inputs in a common voxel format. However, the main spatial interaction descriptor uses only the cross-case consistent P-wave velocity attribute. The geological anomaly score is therefore a $[0, 1]$ low-velocity anomaly score derived from the training active-zone reference distribution of V_p; derived or case-specific attributes such as V_s, modulus, velocity ratio, Poisson's ratio, and density are not used in the main descriptor calculation.

4.3 Diagnostic evaluation metrics

The diagnostic outputs are component–chainage descriptors. For each component c, $A_c(t)$ is a supporting exposure descriptor that reports the sum of geometry weights over candidate relations, and $I_c(t)$ is the main geology-weighted descriptor. The response consistency target is the persistence residual $e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)}$ for each response variable k.

Descriptor–residual association is formally reported with Spearman correlation between $I_c(t)$ and the persistence residual. A threshold sensitivity analysis varies τ_{edge} and $\eta_{\min}$ to check whether the descriptor–residual pattern is tied to one specific graph-construction setting. Null-model comparisons, detrending, circular-shift checks, anomaly-definition sensitivity, and edge-level traceability are applied to the fixed diagnostic pairs. The results are interpreted as chainage-local diagnostic co-variation, not as causal evidence or calibrated operational risk.

5 Results

5.1 Constructed rock–machine interaction graph sequences

The first result is the constructed spatial relation object itself. For each case, TSP-derived rock voxels and parameterized TBM surface nodes are mapped to the same chainage-referenced coordinate system, and the candidate relation set E_t^{rm} is rebuilt at each excavation step. Figure 2 illustrates the construction rule: candidate relations are screened by active-zone membership, distance threshold, surface-normal compatibility, and excavation state.

Table 5 reports the standardized graph support used in all three formal runs. The 3558 rock nodes are the active, not-yet-excavated TSP voxels retained by the common active-zone rule under the 1.0 m voxel grid, $\tau_{\text{zone}} = 5.0$ m, and fixed TBM template. The count is constant because the active-zone window is translated with the TBM and evaluated on a standardized local support, not because the same absolute voxels remain active at all chainages. Because the same parameterized TBM surface, active-zone dimensions, surface sampling resolution, and edge thresholds are applied across cases, the per-step node and edge counts are identical by design. This table is therefore not evidence of dynamic graph-size variation across cases; it documents the fixed spatial support template on which the case-specific q_i, $I_c(t)$, and residual series are computed. The case contrast is expressed by geological attribute values, target chainages, and monitoring residuals, not by different graph-size templates.

Table 5: Standardized graph support template applied in all formal runs. Edge shares are computed from candidate rock–machine relations under the common main graph setting. The identical counts confirm that case contrasts originate from geological attributes and residuals, not from graph-size variation.

Case	Rock nodes	TBM nodes	Candidate edges	Cutterhead share	Shield share
BSLL h=1	3558	1352	14620	0.145	0.855
BSLL h=3	3558	1352	14620	0.145	0.855
SJLS h=3	3558	1352	14620	0.145	0.855

The descriptor statistics in Table 6 explain why SJLS provides clearer diagnostic evidence despite using the same graph support template. SJLS is not stronger because it has a larger graph; it is stronger because the standardized support is populated by a more uniformly high low-velocity anomaly field. This interpretation is conditional on the SJLS geological support being a 1D-extruded TSP velocity profile rather than a fully 3D-inverted voxel field; the SJLS result is therefore evidence of longitudinal descriptor–residual consistency under a common support, not proof of fully 3D component differentiation. Within that boundary, the SJLS interval provides a clearer diagnostic pattern than the compact BSLL intervals.

A notable feature of Table 6 is that the candidate-edge mean anomaly and $I_c(t)$ are numerically very close in all three cases (e.g. BSLL h=1: 0.552 ± 0.157 vs. 0.552 ± 0.158). This near-equality is expected rather than degenerate: $I_c(t)$ aggregates q_i over screened candidate edges, and the geometry weights mainly provide edge-level contribution traceability in these fixed pairs. The diagnostic value shown here does not lie in a different central tendency from the unweighted candidate-edge mean. It lies in retaining the component-specific, chainage-specific candidate-edge aggregation that the pooled active-zone mean conceals. The null-model comparison in Section 5.4 tests this directly by comparing the proposed descriptor against the candidate-edge mean (uniform-edge variant) and the global active-zone mean (global-Vp variant).

Table 6: Case-level descriptor attribute statistics. Mean anomaly is the candidate-edge mean of the bounded low-velocity score q_i; $I_c(t)$ is the candidate-edge anomaly descriptor pooled over components and test chainages.

Case	Mean anomaly score	$I_c(t)$	$A_c(t)$
BSLL h=1	0.552 ± 0.157	0.552 ± 0.158	1170.8 ± 296.9
BSLL h=3	0.537 ± 0.129	0.537 ± 0.130	1170.9 ± 297.4
SJLS h=3	0.913 ± 0.045	0.913 ± 0.045	1170.9 ± 294.2

Table 7 summarizes the retained component–chainage structure directly. BSLL shows stronger cross-component contrast in $I_c(t)$ but has too few test samples to support a stable residual association. SJLS shows weaker cross-component contrast, as expected for a 1D-extruded velocity profile with common cross-sectional support; its stronger evidence therefore comes from longitudinal descriptor–residual co-variation, not from fully 3D component differentiation.

Table 7: Component contrast in the $I_c(t)$ descriptor field. The statistics summarize the per-component, per-chainage descriptor matrix and are not response-model scores.

Case	Component-mean range	Mean within-component SD	Full $I_c(t)$ range
BSLL h=1	0.220	0.132	0.570
BSLL h=3	0.211	0.102	0.426
SJLS h=3	0.016	0.044	0.150

5.2 Component–chainage descriptor patterns and residual traces

The main graph-derived output is the component–chainage descriptor map built from $I_c(t)$. In each case, cutterhead, front-shield, middle-shield, and tail-shield values are computed across the test chainage interval as TSP anomaly summaries over screened candidate rock–machine relations.

Figure 3 previews the per-component $I_c(t)$ series across the three case runs on the common descriptor support. The BSLL cases operate over compact chainage intervals (about 6–7 m), while SJLS spans a longer test interval (about 16 m) with a markedly higher anomaly level. This figure motivates the case-level diagnostic contrast examined in the remainder of this section.

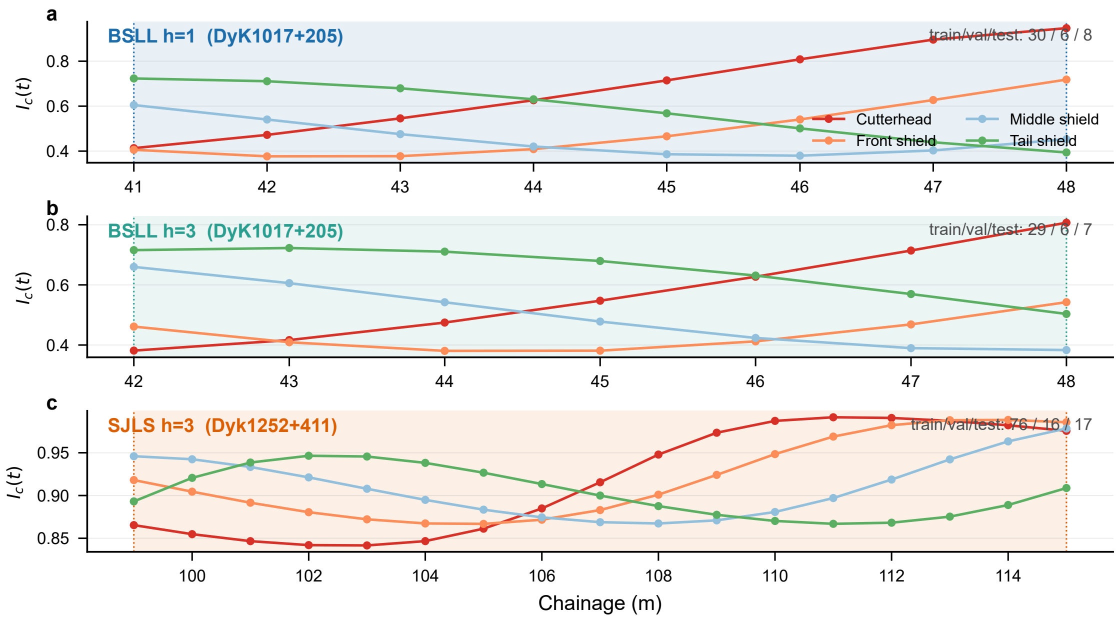

Figure 3: Per-component $I_c(t)$ descriptor series for the three formal case runs on the common standardized graph support. Shaded bands mark the test chainage interval; BSLL cases are compact ($n = 8, 7$), while SJLS provides a longer test interval ($n = 17$) with a uniformly high low-velocity anomaly level.

Figure 4 links these descriptor maps to the fixed diagnostic residual traces. Each case panel holds the diagnostic component and response fixed, showing the selected $I_c(t)$ series above the corresponding persistence residual. This layout makes the evidence chain explicit: the graph first produces a component–chainage descriptor, and the residual trace then tests whether that descriptor co-varies with response changes left after persistence.

5.3 Representative diagnostic descriptor–residual associations

The formal diagnostic statistic is Spearman correlation between $I_c(t)$ and the persistence residual $e_{t+h}^{(k)}$. Table 8 reports the representative diagnostic pairs together with raw Spearman correlation, linearly detrended Spearman correlation, and the circular-shift diagnostic. The pairs are representative diagnostic slices selected from the main descriptor–response matrix before the secondary null, alignment, and sensitivity checks. They cover the compact BSLL h=1 interval, the BSLL h=3 shield-pressure interval, and the external SJLS shield-pressure case, while retaining component–response combinations with direct engineering plausibility. They are not claimed to be unique physically responsible components or globally strongest pairs; Appendix A reports the full exploratory matrix without multiplicity adjustment. The values are descriptive and are not treated as confirmatory hypothesis tests.

Table 8: Representative diagnostic descriptor–residual diagnostics for fixed component–response pairs. Bootstrap intervals are descriptive percentile intervals for detrended Spearman correlations (10,000 resamples).

Case	Diagnostic pair	n	Raw ρ	Detrended ρ	95% CI	Shift check
BSLL h=1	Front shield–advance rate	8	0.381	0.214	$[-0.671, 1.000]$	0.375
BSLL h=3	Front shield–shield pressure	7	-0.750	-0.464	$[-0.957, 0.769]$	0.429
SJLS h=3	Cutterhead–shield pressure	17	-0.912	-0.772	$[-0.968, -0.375]$	0.059

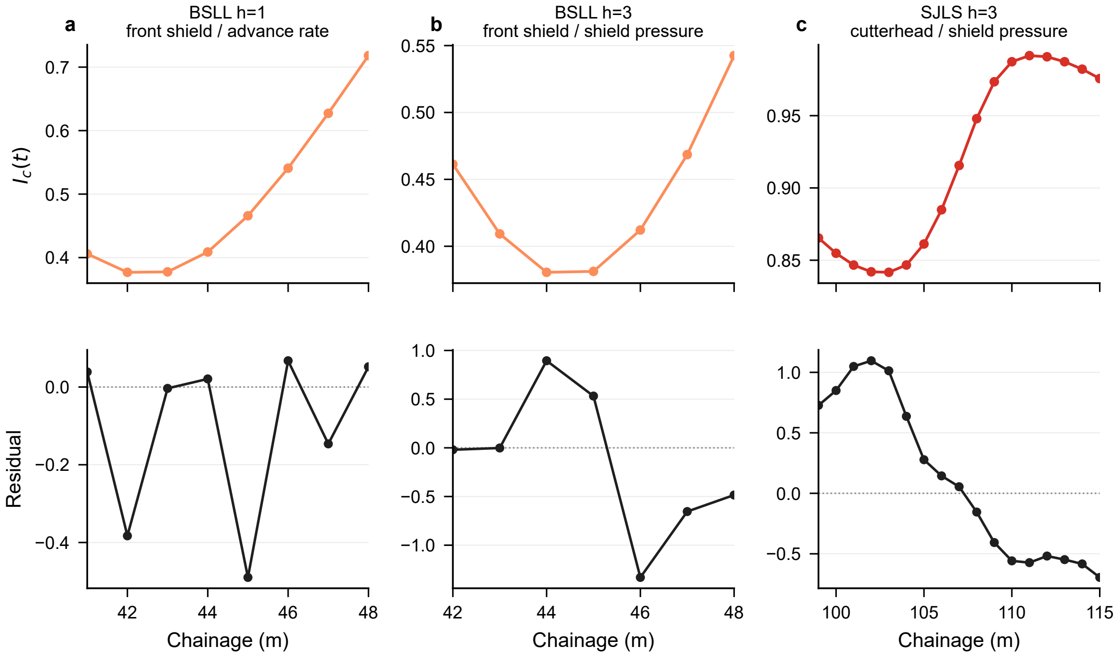

Figure 4: Fixed diagnostic descriptor–residual traces. Upper panels show the selected component $I_c(t)$ series for each formal case; lower panels show the matching persistence residual series in the standardized monitoring scale used for residual computation.

The bootstrap intervals are wide for the two BSLL cases, reflecting the small test intervals ($n = 8$ and $n = 7$). SJLS shows the most evident descriptor–residual co-variation. Its cutterhead $I_c(t)$–shield-pressure residual association remains strong after linear detrending ($\rho = -0.772$), whereas the BSLL evidence weakens under detrending. This difference is consistent with the longer SJLS test interval and clearer external TSP velocity contrast. The BSLL results are retained as compact diagnostic checks, but they do not support a strong case-general rule.

For completeness, Appendix A reports the full component–response Spearman matrix for $I_c(t)$. The matrix is exploratory and unadjusted for multiple testing. The fixed pairs in Table 8 are diagnostic slices interpreted together with the null, detrending, sensitivity, and traceability checks below.

5.4 Descriptor-level null comparisons

The null-model comparison tests whether the fixed diagnostic associations can be explained by simpler summaries. Figure 5 shows raw and detrended correlations for the proposed descriptor and four descriptor-level null variants. The most important diagnostic is the detrended comparison, because chainage and global anomaly can both carry strong monotonic trends in short test intervals. Table 9 lists the detrended correlations for the proposed descriptor and four descriptor-level null variants.

Table 9: Detrended Spearman correlations for the proposed descriptor and four descriptor-level null variants. The component-label disruption diagnostic is reported separately in Table 10.

Case	Proposed	Chainage	Global Vp	Distance only	Uniform edge
BSLL h=1	0.214	0.000	-0.190	0.000	0.214
BSLL h=3	-0.464	-0.145	-0.321	0.000	-0.464
SJLS h=3	-0.772	-0.009	-0.179	0.000	-0.772

For SJLS, the raw chainage-only and global-Vp associations are high, but both collapse after

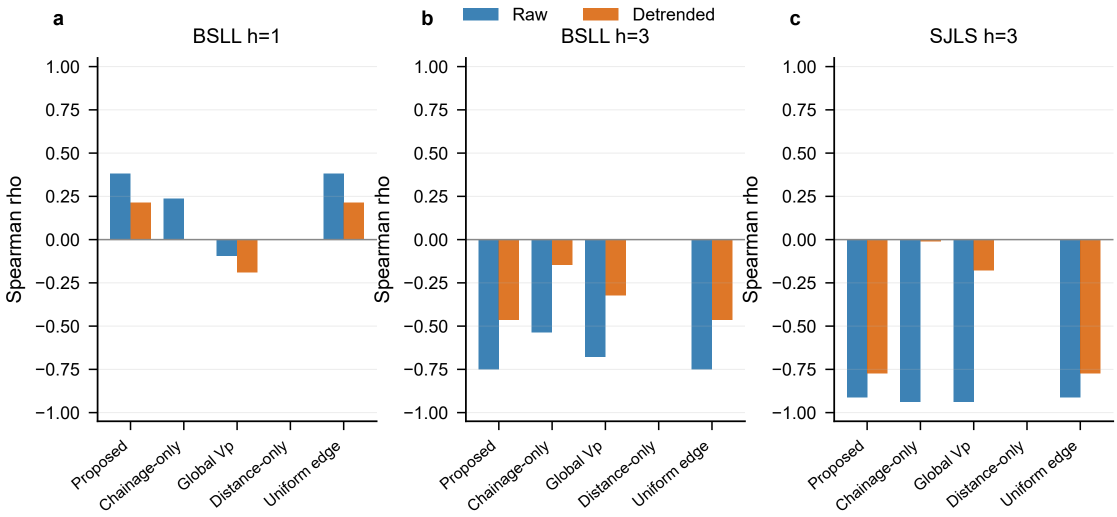

Figure 5: Descriptor-level null-model comparison for the representative diagnostic pairs. Bars report raw and linearly detrended Spearman correlations for the proposed descriptor and four descriptor-level null variants. For the fixed pairs, the uniform-edge descriptor is essentially the same as the proposed descriptor, so this figure does not demonstrate an independent performance benefit of distance–normal geometry weighting. The comparison evaluates descriptor diagnostic structure, not prediction accuracy.

detrending. The proposed descriptor retains a strong detrended association, indicating that the component-specific candidate-edge aggregation contains more than a simple chainage trend. Because the uniform-edge and geometry-weighted descriptors are nearly identical in these fixed pairs, this comparison does not isolate an independent gain from distance–normal geometry weighting.

A step-wise component-label disruption diagnostic ($N = 1000$ permutations, with component labels independently shuffled at each chainage step and both the observed and disrupted series linearly detrended against chainage before correlation) evaluates whether the fixed component label matters after the common graph support and chainage trend are removed. As reported in Table 10, the descriptive permutation proportions are 0.588 (BSLL h=1), 0.325 (BSLL h=3), and < 0.001 (SJLS h=3). This diagnostic is a label-disruption check rather than a block-wise or case-general permutation test. It can overstate component specificity because the relabelling is performed independently at each chainage step and therefore does not preserve complete component series. The values are therefore read only as descriptive disruption diagnostics.

Table 10: Step-wise component-label disruption diagnostic on detrended series. At each chainage step, component labels are independently shuffled ($N = 1000$ permutations); both observed and permuted descriptor series are linearly detrended against chainage before computing Spearman correlation with the detrended residual. The reported proportion is descriptive and unadjusted.

Case	Component	n	Observed ρ	Permutation p
BSLL h=1	front shield	8	0.214	0.588
BSLL h=3	front shield	7	−0.464	0.325
SJLS h=3	cutterhead	17	−0.772	< 0.001

Table 11 reports a stricter whole-series component reassignment diagnostic. Instead of shuffling labels at each chainage, it compares the target component's complete detrended $I_c(t)$

series with the complete series from the other components for the same residual. This check shows why the component-specific interpretation must remain cautious: the fixed target is not always the strongest component series, especially for SJLS where tail-shield has a larger absolute detrended association. The result supports descriptor–residual consistency more strongly than unique component-label specificity. Therefore, SJLS is treated as the strongest longitudinal residual-consistency case, whereas BSLL remains the only case with fully 3D TSP voxel support for component differentiation, although its short test interval limits residual evidence.

Table 11: Whole-series component reassignment diagnostic. Values are detrended Spearman correlations between each complete component $I_c(t)$ series and the fixed residual; no per-step label shuffling is used.

Case	Target component	Target ρ	Other ρ range	Strongest other ρ	Interpretation
BSLL h=1	front shield	0.214	−0.214 to 0.167	−0.214	tied absolute strength
BSLL h=3	front shield	−0.464	−0.500 to 0.464	−0.500	target not strongest
SJLS h=3	cutterhead	−0.772	−0.135 to 0.877	0.877	target not strongest

The null variants separate three questions. The global-Vp variant asks whether a case-level active-zone mean is sufficient. The distance-only variant asks whether geometric exposure alone explains the residual association. The uniform-edge variant keeps the fixed component and the screened candidate-edge set, but removes distance–normal weights by using an unweighted mean of q_i. In Table 9, the uniform-edge values are numerically the same as the proposed descriptor for the fixed pairs. The current evidence therefore supports component-specific candidate-edge aggregation over global active-zone averaging, but it does not show a separate incremental benefit of geometry weighting within these samples. The geometry weights remain part of the representation because they define a transparent, queryable edge contribution $w_{ij}^{rm}(t)q_i$, but their independent empirical gain requires larger or more spatially heterogeneous cases.

5.5 Sensitivity to graph thresholds and anomaly definitions

Figure 6 tests whether the fixed $I_c(t)$–residual pairs are preserved when the edge-distance threshold τ_{edge} and normal-compatibility threshold $\eta_{\min}$ are varied. The same descriptor–response pair is held fixed in each case, so the figure evaluates threshold stability rather than searching for a new strongest pair.

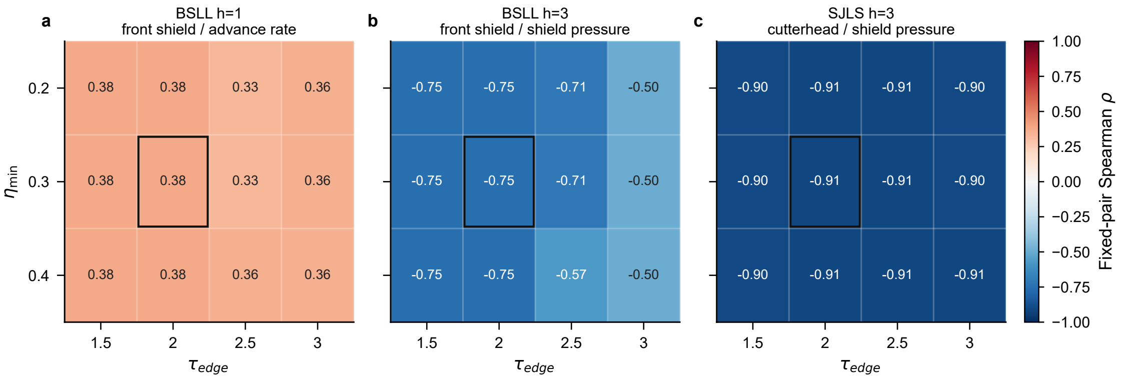

Figure 6: Sensitivity of fixed $I_c(t)$–residual pairs to graph-construction thresholds. Each panel reports raw Spearman ρ while varying τ_{edge} and $\eta_{\min}$ for one formal case; the descriptor–response pair is held fixed throughout.

Table 12 reports the anomaly-definition sensitivity for the same fixed diagnostic pairs. SJLS remains strong under Vp, Vs, and mean(Vp, Vs) low-anomaly definitions after detrending. BSLL is more sensitive to the anomaly definition, which is consistent with its compact test intervals.

Table 12: Detrended Spearman correlations under alternative anomaly-score definitions for the fixed diagnostic pairs.

Case	Vp main	Vs low	Vp/Vs low	Mean(Vp, Vs)
BSLL h=1	0.214	-0.214	0.214	-0.024
BSLL h=3	-0.464	0.179	-0.179	-0.643
SJLS h=3	-0.772	-0.873	0.338	-0.863

Table 13 reports the TSP–TBM chainage alignment uncertainty check. The SJLS association remains negative under ± 1 m offsets and remains strong after detrending at the 0 and $+1$ m settings. The BSLL checks are more variable, again reflecting their short test intervals.

Table 13: Sensitivity of fixed diagnostic pairs to ± 1 m longitudinal descriptor alignment offsets. Values report raw/detrended Spearman correlations.

Case	-1 m	0 m	$+1$ m
BSLL h=1	0.250 / -0.107	0.381 / 0.214	-0.071 / 0.250
BSLL h=3	-0.143 / 0.257	-0.750 / -0.464	-0.657 / -0.714
SJLS h=3	-0.850 / -0.397	-0.912 / -0.772	-0.938 / -0.874

5.6 Edge-level traceability

The graph representation also supports edge-level inspection. For the SJLS diagnostic pair, Figure 7 shows the top contributing candidate rock–machine relations at the selected chainage. Each contribution is traceable to a rock voxel, a TBM surface node, a distance, a normal compatibility value, a geometry weight, and a low-velocity anomaly score; the figure displays the leading spatial edges and their ranked contributions, while the full edge-attribute rows remain available in the diagnostic outputs.

This traceability is the main practical difference between the proposed graph descriptor and a monitoring-only residual curve. The descriptor can be queried back to the spatial relation set that produced it, while the residual curve alone cannot identify which geological voxels and machine components were involved in the diagnostic summary.

6 Discussion

6.1 Spatial data-model contribution

The main contribution is not a higher-accuracy response predictor, but a chainage-referenced spatial relation model that makes the otherwise implicit rock–machine relation set explicit, inspectable, and queryable. Monitoring sequences can describe response inertia, and TSP voxels can describe geological heterogeneity, but neither representation alone states which geological voxels are candidate interaction partners for which TBM components at a given chainage. The proposed graph closes this representational gap by linking rock voxels, TBM surface nodes, geometry-compatible candidate edges, and monitoring residuals in one excavation coordinate system.

This distinction matters for spatial modelling because the graph is introduced as an explicit data representation rather than as a predictive architecture. It has defined entities, relation types, screening rules, edge weights, component aggregation, and traceability back to individual candidate relations. The descriptor results are therefore interpreted as evidence that the representation can organise rock–machine relations into component–chainage diagnostic summaries.

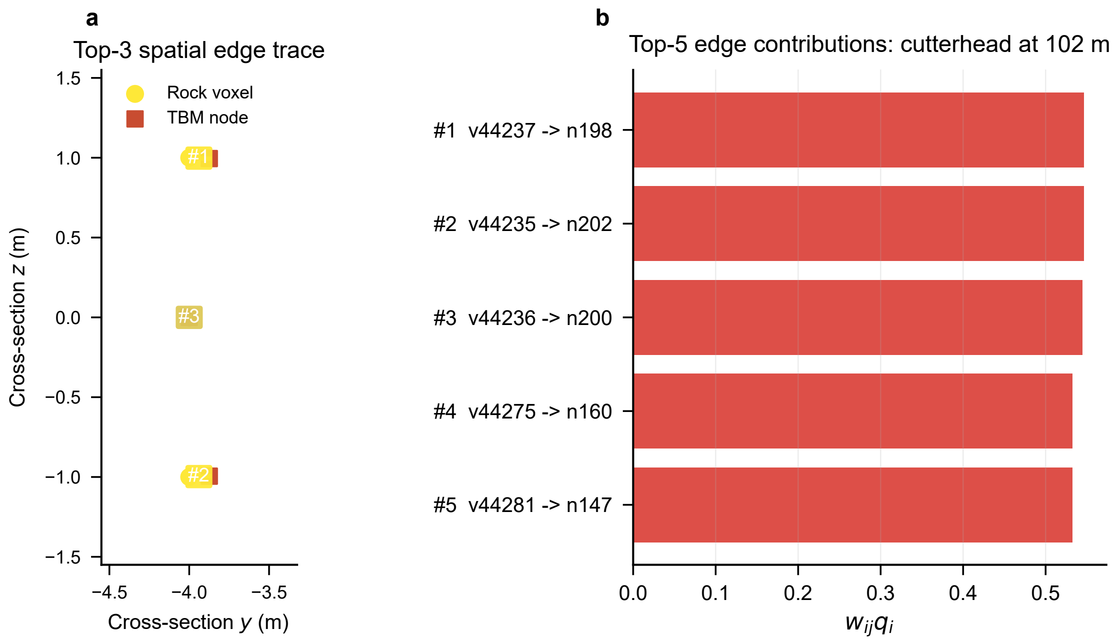

Figure 7: Traceability example for a fixed diagnostic pair. Candidate rock–machine edges are ranked by $w_{ij}^{rm}(t)q_i$; the left panel maps the top three spatial edges in cross-section, and the right panel reports the top five weighted contributions.

From a digital-earth perspective, the contribution is an underground spatial data modelling layer for a moving construction process. It turns TSP-derived geological support, TBM surface geometry, and chainage-indexed monitoring into dynamic spatial entities and relations that can be inspected and queried. This layer is compatible with digital twin workflows, but it addresses a narrower representational problem: how to make the rock–machine relation support explicit before any prediction, control, or risk-calibration module is built.

6.2 Diagnostic meaning beyond monitoring curves

Comparing $I_c(t)$ with persistence residuals reframes the evaluation question from "does the graph beat persistence?" to "does the graph encode spatial diagnostic information consistent with the response changes left after persistence?" This is a more defensible question for the present data scale. The SJLS results illustrate the value of this framing: raw chainage and global Vp summaries are strongly associated with the selected residual, but their associations largely disappear after linear detrending, whereas the proposed candidate-edge descriptor remains strong.

The BSLL results are more limited. They show that the graph and descriptors can be constructed consistently in compact intervals, but the residual-consistency evidence weakens after detrending and changes with anomaly definition. This does not invalidate the graph construction; rather, it defines the appropriate claim boundary. The evidence supports a diagnostic spatial-representation claim, not a universal component–response law.

6.3 Case-specific evidential strength

The contrast between SJLS and BSLL is not treated as a success–failure dichotomy. It shows that descriptor usefulness depends on geological contrast, monitoring quality, operational stability, and the length of the evaluated chainage interval. SJLS contains a longer test interval and a clearer external TSP velocity contrast, which makes descriptor–residual co-variation more visible. BSLL h=1 and h=3 contain only 8 and 7 test samples, respectively, and their residuals may be more strongly affected by operational control and local construction variability.

The circular-shift checks are also read conservatively. They are useful as small-sample diagnostics for monotonic sequence structure, but they are not formal multiplicity-adjusted tests. The most coherent support comes from the combined pattern: fixed diagnostic pairs, detrended comparison, descriptor-level null models, threshold sensitivity, anomaly-definition sensitivity, and edge-level traceability.

The step-wise component-label disruption diagnostic (Table 10) is retained only as a label-disruption check. Because it shuffles labels independently at each chainage step, it does not preserve complete component trajectories and should not be read as proof that physically different TBM components are exchangeable. The whole-series component reassignment diagnostic (Table 11) is more conservative and shows that the fixed target component is not always the strongest complete component series. The evidence therefore supports candidate-edge descriptor consistency more strongly than unique component-label specificity.

6.4 Interpretation boundary

The proposed descriptors are summaries of screened TSP-derived anomalies around TBM components, with geometry weights retained for edge-level traceability. They are not measured contact force, contact pressure, jamming probability, causal attribution, or calibrated operational risk. This boundary is important because the graph encodes candidate spatial relations, not observed physical contact. Independent engineering labels would be required before the descriptors could be interpreted as risk probabilities or contact-force estimates.

A second boundary concerns the standardized graph support. All three formal runs use an identical parameterized TBM surface, active-zone geometry, surface sampling resolution, and edge thresholds, so the per-step node and edge counts are the same across cases by construction. This standardization keeps the descriptor comparable across cases and prevents graph-size variation from being confounded with geological contrast, but it also means that the current framework cannot test whether a different TBM geometry, a coarser surface sampling, or a different active-zone extent would change the diagnostic pattern. The standardized support is therefore both a comparability device and a scope limitation: the reported evidence is conditional on one graph template, and sensitivity to machine geometry and active-zone dimensions remains for future work.

6.5 Limitations and future work

Several limitations remain. First, the current evidence comes from two tunnel cases and short diagnostic test intervals: BSLL contains only 8 and 7 test samples for h=1 and h=3, respectively, and SJLS contains 17 test samples. The reported associations are therefore descriptive diagnostics rather than case-general component–response laws. Second, TSP inversion uncertainty is not propagated through the voxel graph; future work should attach uncertainty to voxel attributes and evaluate how inversion and spatial resolution affect candidate relation sets. For the SJLS case in particular, the geological support is a 1D-extruded velocity profile rather than a fully 3D-inverted voxel field, so cross-sectional geological variation is not captured by the current descriptor.

Third, the step-wise component-label disruption diagnostic only disrupts labels within the standardized graph support. Larger datasets would allow stronger null designs and inference that accounts for multiple comparisons. Fourth, the study lacks independent engineering labels such as jamming records, shield pressure anomaly logs, construction intervention logs, or geological exposure records. Such evidence would be required before the descriptors could be validated as operational risk indicators. Finally, the main descriptor uses a transparent Vp-based low-velocity anomaly score. Multi-attribute scores using Vs, modulus, Vp/Vs, Poisson's ratio, or density require explicit weighting rules or independent validation before they can replace the main descriptor.

7 Conclusion

This study develops a chainage-referenced spatial entity-relation framework for rock–TBM interaction representation. The framework places TSP-derived rock voxels, parameterized TBM surface components, and monitoring responses in a common excavation coordinate system, so that geological units, machine components, and response residuals can be analysed through explicit spatial relations.

The framework constructs geometry-constrained candidate rock–machine relations using active-zone membership, distance decay, surface-normal compatibility, and excavation-state constraints. It then derives component–chainage diagnostic descriptors by aggregating TSP anomaly scores over screened candidate relations, retaining geometry weights for edge-level traceability. Descriptor-level null comparisons, detrending, circular-shift checks, threshold sensitivity, anomaly-definition sensitivity, component-series reassignment, and edge-level traceability are used to evaluate whether these descriptors contain residual-consistent diagnostic information.

The two case studies indicate that the descriptors can organise rock–machine relations into interpretable component–chainage patterns and, most clearly in the SJLS case, retain residual-consistent co-variation after simple trend-based null explanations are considered. The evidence supports a diagnostic spatial representation claim rather than contact-force recovery, calibrated jamming risk, or broad predictive dominance. The proposed framework provides a spatial representation basis for future rock–machine interaction analysis and digital-twin-oriented TBM excavation analysis.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The raw TBM monitoring and TSP data are project data and are not publicly available because they contain engineering and site information. The materials that can be shared are the experiment configurations, analysis scripts, figure scripts, and derived result summaries used to reproduce the reported tables and figures. These materials are available from the corresponding author upon reasonable request, subject to project data-use permissions; the raw site data are not redistributed. To support reproducibility of the spatial graph construction without disclosing site data, the authors can also provide a synthetic demonstration package containing a small voxel field, TBM surface nodes, candidate edge lists, descriptor calculation scripts, and residual diagnostic scripts.

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A Exploratory descriptor–response matrix

Table 14 reports the full unadjusted component–response Spearman matrix for all cases, included here for completeness. These values are exploratory descriptor diagnostics and are not adjusted for multiple testing.

Table 14: Full unadjusted component–response Spearman matrix for $I_c(t)$–persistence-residual association. Correlations are exploratory descriptor diagnostics and are not adjusted for multiple testing.

Case	Component	Adv. rate	Torque	Thrust	Penetr.	Shield press.
BSLL h=1	Cutterhead	0.238	-0.286	0.048	-0.095	0.095
BSLL h=1	Front shield	0.381	-0.429	0.190	0.095	-0.095
BSLL h=1	Middle shield	-0.048	0.262	0.452	-0.167	0.143
BSLL h=1	Tail shield	-0.238	0.286	-0.048	0.095	-0.095
BSLL h=3	Cutterhead	-0.214	-0.714	-0.036	-0.214	-0.536
BSLL h=3	Front shield	0.286	-0.179	0.143	0.250	-0.750
BSLL h=3	Middle shield	0.214	0.714	0.036	0.214	0.536
BSLL h=3	Tail shield	0.179	0.750	0.107	0.143	0.571
SJLS h=3	Cutterhead	0.860	0.446	-0.669	0.669	-0.912
SJLS h=3	Front shield	0.640	0.152	-0.350	0.475	-0.745
SJLS h=3	Middle shield	-0.225	-0.449	0.478	-0.282	-0.042
SJLS h=3	Tail shield	-0.846	-0.324	0.733	-0.811	0.784